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Biosensors exploit the molecular recognition properties that are unique to biomolecules in order to provide highly specific assays for compounds in complex mixtures. Many common biosensor designs are prone to significant mass transport limitations that lead to prolonged sensor response times. As a result, such biosensors have limited use in real-time monitoring of dynamic processes and can be difficult to rinse and re-use.
This research aims to accelerate mass transport in barrier regions that occur in common biosensor designs. For biosensors based on gel-entrapped enzymes, we have developed an internal pumping method, wherein charged silica nanoparticles are entrapped in the gel, and application of an external electric field drives electroosmotic fluid flow around the nanoparticles. Mixing within the gel is driven by applying time varying quadrupolar electric fields that create temporally and spatially varying electroosmotic laminar flows. We first demonstrated the feasibility of using charged nanoparticle inclusions inside polyacrylamide gels in order to pump electroosmotic flows. This was demonstrated with DC fields that produced several-fold greater tracer fluxes across a nanoparticle-laden gel slab compared to diffusion and

Task Description:	compared to tracer fluxes when DC fields were applied in the absence of the charged nanoparticle inclusions. Thus electroosmotic fluxes were stronger than any diffusive or electrophoretic fluxes. Subsequently, tracer mixing within gels has been successfully demonstrated in response to quadrupolar fields applied across silica nanoparticle-laden gels. This was demonstrated using fluorescence recovery after photobleaching to measure the mixing of bleached and unbleached fluorescent tracers in the gel. Ten- to 100-fold reductions in mixing times have been achieved relative to diffusive mixing. In addition, interfacial micromixing strategies were investigated in order to promote mass transfer near solid surfaces. Electrohydrodynamic flows, combined with dielectrophoretic forces, were used to organize colloids and cells near both uniform and micropatterned electrode surfaces. Theories were developed to predict how electric field frequency and electrode composition and geometry dictate the shape and magnitude of the nonuniform electric fields that drive the interfacial flows, and to predict the induced dipole strength of particles in ac electric fields. This dictates the sign and magnitude of a particle's migration in response to the nonuniform fields. As a further elaboration on this concept, a microfluidic technique was developed to laterally separate mixtures of cells and/or colloids based on their dielectric properties. This technique coupled ac electroosmosis with dielectrophoresis to drive separations faster than was possible by dielectrophoresis alone, and it enabled separations that were not possible with dielectrophoresis alone. The objective of this research program was to develop electrokinetic techniques that minimize mass transport limitations in biosensors and thereby improve their signal response times. This is intended to expand the applicability of biosensors to dynamic processes with short timescales. This research has developed techniques to increase the speed at which biosensors detect molecules in solution. Biosensors contribute significantly to the approximately $19 billion world market for medical diagnostics. Biosensors use biological molecules to detect trace amounts of specific chemical compounds in complicated chemical mixtures, for example, detecting hormones or sugars in blood, or toxins in groundwater or processed foods. A major limitation of biosensors is that they often require long times to produce a stable signal, referred to as a long response time. The long response time problem prevents biosensors from being used in any situation where chemical concentrations change rapidly, for example preventing one from using biosensors to monitor rapidly changing product concentrations in a pharmaceutical manufacturing process. Furthermore, at a time when combinatorial chemistry can generate thousands of new molecules to be tested for pharmaceutical activity, there is a need for rapid analysis, or high throughput screening, of the biological effectiveness of those molecules. Biosensors are ideally suited for measuring biological effects, but the long response time problem significantly limits their use in a high throughput screening mode. The results of this research program are expected to expand the economic market for biosensors and to accelerate the screening of new pharmaceuticals in order to deliver new therapies to the general population. Furthermore, new strategies developed in this project to separate cells in microfluidic devices have the potential to lead to inexpensive, portable devices for field analysis of biological samples, or to inexpensively analyze medical samples for cell type and abundance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Two female PhD students, one of whom, a Hispanic woman, is a member of an under-represented minority in science and engineering practice, were directly supported by this research project. Research results are providing new understanding of convection and particle motion in uniform as well as nonuniform electric fields. It is also providing new theoretical understanding of the polarization of dielectric particles in electric fields and the frequency dependence of electric fields in electrolyte solutions. This will be broadly applicable to electrochemical and electrophoretic technologies.
Task Progress:	1. Internal Pumping. Protocols for the reproducible preparation of silica nanoparticle-laden gels were developed, leading to highly reproducible results on the enhancement of solute fluxes across polyacrylamide gels via electroosmotic internal pumping. The uniformity of the charged inclusion distribution within the gels was confirmed by small angle neutron scattering. The effects of applied electric field strength, particle concentration and particle size on the degree of mass transfer enhancement were measured systematically. The enhancement scales with the total surface area of embedded particles, as expected for a surface-driven flow process. Using dc fields aligned parallel to the solute concentration gradient, solute fluxes are enhanced by electroosmosis. For example, at a particle load of 2 vol%, solute flux is quadrupled by application of a 3 mA current across the gel. If the field is applied antiparallel to the solute concentration gradient, it is possible to drive an electroosmotic flow that runs counter to diffusion and prevents solute flux across the gel. We anticipate that this may become useful as an electrically controlled microfluidic valve or as an electrically controlled drug delivery technology. For example, medication could be released on demand from an implanted gel reservoir when needed, in response to an external electrical signal. Having demonstrated the feasibility of internal pumping with dc currents, fluorescence recovery after pattern photobleaching experiments were performed to test the enhancement of intra-gel mixing by applying time varying quadrupolar electric fields to nanoparticle-laden gels. Applied field waveforms were designed to drive laminar mixing by stretching and wrapping fluid elements within the gel. The image of a candy mixer is an appropriate analogy. 10 to 100-fold reductions in mixing times have been achieved. 2. Interfacial Micromixing and Particle Manipulation. We have demonstrated that cells and particles can be differentially manipulated and positioned adjacent to a microelectrode array when an ac field is applied between the array and a uniform electrode oriented parallel to the array. Cells and particles migrate in response to a coupling between ac electroosmosis and dielectrophoresis. Thus, the separation is based on differences in cell or particle polarizability. This has been exploited to efficiently separate different particle types at low frequencies (~10 to 1000 Hz) and low electric field strengths (~ 10 V/cm) that are appropriate for biological applications. Separations were first demonstrated in a model system consisting of yeast and colloidal polystyrene. Yeast has also been separated from red blood cells using this technique. This experimental work has been complemented by theoretical analysis. We have developed a new model of particle polarizability derived from the full electrokinetic theory, as well as a new model of frequency-dependent electrode impedance and the structure of ac electric fields in conducting media. These theoretical analyses indicate that cells with small differences in dielectric properties can be separated by judiciously specifying the conductivity of the suspending medium. This variable is key because it not only affects how the difference between two cells' polarizabilities varies with frequency, but it also controls the electric field structure around the microelectrodes. We anticipate that cells with small differences in membrane conductivity, perhaps because of a disease state, could be separated by this technique. We have also demonstrated the ability to position vesicles using this technique. Vesicles can

	be steered by dielctrophoresis and ac electroosmosis and subsequently ruptured by the electric field, providing a high degree of spatial control over vesicle content release behaviors. Vesicles can be loaded with reactants, and by differentially manipulating different reactant-laden vesicles or colloidal particles, we can control when and where reactions occur in microfludic devices.
Bibliography Type:	Description: (Last Updated: 08/30/2006)
Articles in Peer-reviewed Journals	Zhou H, White LR, Tilton RD. "Lateral separation of colloids or cells by dielectrophoresis augmented by AC electroosmosis." J Colloid Interface Sci. 2005 May 1;285(1):179-91. PMID: 15797412 , May-2005
Articles in Peer-reviewed Journals	Zhou H, Preston MA, Tilton RD, White LR. "Calculation of the electric polarizability of a charged spherical dielectric particle by the theory of colloidal electrokinetics." J Colloid Interface Sci. 2005 May 15;285(2):845-56. PMID: 15837504 , May-2005
Articles in Peer-reviewed Journals	Zhou H, Preston MA, Tilton RD, White LR. "Calculation of the dynamic impedance of the double layer on a planar electrode by the theory of electrokinetics." J Colloid Interface Sci. 2005 Dec 1;292(1):277-89. PMID: 15996678 , Dec-2005
Articles in Peer-reviewed Journals	Zhou H, Tilton RD, White LR. "The role of electrode impedance and electrode geometry in the design of microelectrode systems." J Colloid Interface Sci. 2006 May 15;297(2):819-31. PMID: 16332373 , May-2006
Articles in Peer-reviewed Journals	Zhou H, White LR, Tilton RD. "Microphase separation during binary electrophoretic deposition of particles with dissimilar polarizabilities." Colloids and Surfaces A: Physicochemical and Engineering Aspects, 2006 Apr 5;277(1-3):119-30. , Apr-2006
Articles in Peer-reviewed Journals	Matos MA, White LR, Tilton RD. "Electroosmotically enhanced mass transfer through nanoparticle-doped polyacrylamide gels." J Colloid Interface Sci. 2006 Aug 1;300(1):429-36. Epub 2006 Mar 18. PMID: 16603176 , Aug-2006
Articles in Peer-reviewed Journals	Matos MA, White LR, Tilton RD. "Mixing enhancement within polyacrylamide gels via embedded nanoparticle electroosmotic pumps." manuscript in preparation , Feb-2006